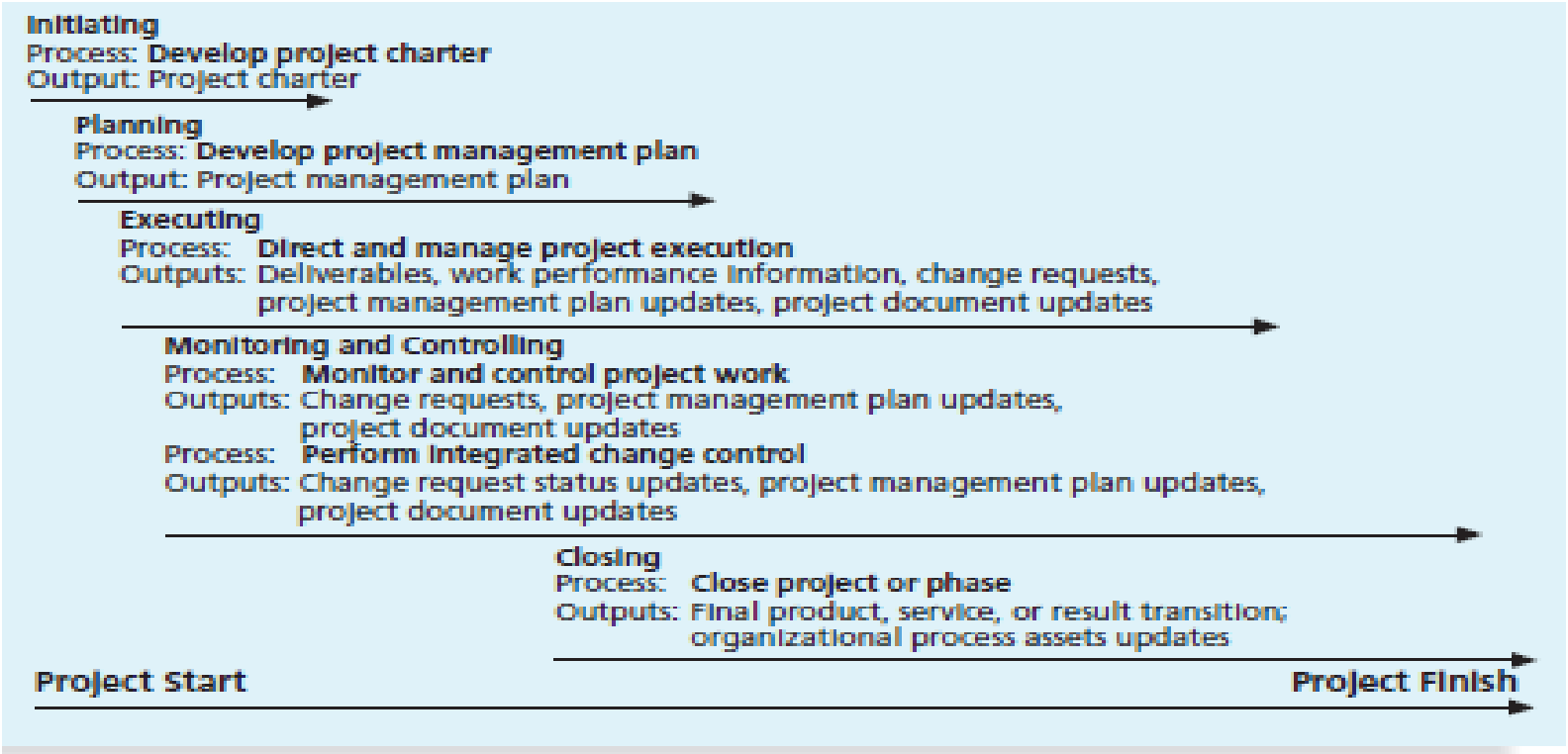
Project Initiation & Integration

Topic # 3

Chapter 4 – Schwalbe

Project Integration Management

- **Project integration management** involves coordinating all of the other project management knowledge areas throughout a project s life cycle.
- Project Integration Management Summary:



Methods for Selecting Projects

- Focusing on broad organizational needs
- Categorizing IT projects (Problems, Opportunities and directives)
- Performing net present value or other financial analysis
- Using a weighted scoring model
- Implementing a balanced scorecard

Financial Considerations for Selecting Projects

- Three primary methods for determining the projected financial value of projects include
  - Net Present Value (NPV),
  - Return on investment (ROI) and
  - Payback period.

Net Present Value (NPV)

- **NPV** is a method of calculating the expected net monetary gain or loss from a project by discounting all expected future cash inflows and outflows to the present point in time.
- The time value of money recognizes that a dollar today is worth more than a dollar one year from now.

Payback Analysis

- **Payback analysis** – a technique for determining if and when an investment will pay for itself.
- **Payback period** – the period of time that will lapse before accrued benefits overtake accrued and continuing costs.

Present Value Formula

Present value – the current value of a dollar at any time in the future.

$$PV_n = 1/(1 + i)^n$$

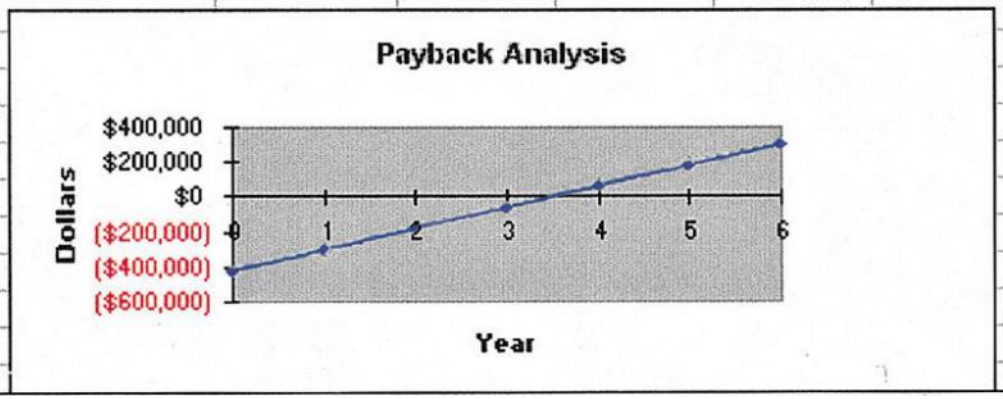
Where ***n*** is the number of years and **i** is the discount rate.

Discount rate – a percentage similar to interest rates that you earn on your savings.

In most cases the discount rate for a business is the **opportunity cost** of being able to invest money in other projects or investments

Payback Analysis for a Project

	A	B	C	D	E	F	G	H	I
4	Cash flow description	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	
5	Development cost:	($418,040)							
6	Operation & maintenance cost:		($15,045)	($16,000)	($17,000)	($18,000)	($19,000)	($20,000)	
7	Discount factors for 12%:	1.000	0.893	0.797	0.712	0.636	0.567	0.507	
8	Time-adjusted costs (adjusted to present value):	($418,040)	($13,435)	($12,752)	($12,104)	($11,448)	($10,773)	($10,140)	
9	Cumulative time-adjusted costs over lifetime:	($418,040)	($431,475)	($444,227)	($456,331)	($467,779)	($478,552)	($488,692)	
10									
11	Benefits derived from operation of new system:	$0	$150,000	$170,000	$190,000	$210,000	$230,000	$250,000	
12	Discount factors for 12%:	1.000	$0.893	$0.797	$0.712	$0.636	$0.567	$0.507	
13	Time-adjusted benefits (current of present value):	$0	$133,950	$135,490	$135,280	$133,560	$130,410	$126,750	
14	Cumulative time-adjusted benefits over lifetime:	$0	$133,950	$269,440	$404,720	$538,280	$668,690	$795,440	
15		0	1	2	3	4	5	6	
16	Cumulative lifetime time-adjusted costs + benefits:	($418,040)	($297,525)	($174,787)	($51,611)	$70,501	$190,138	$306,748	
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Return-on-Investment Analysis (ROI)

Return-on-Investment (ROA) analysis – a technique that compares the lifetime profitability of alternative solutions.

> The ROI for a solution or project is a percentage rate that measures the relationship between the amount the business gets back from an investment and the amount invested.

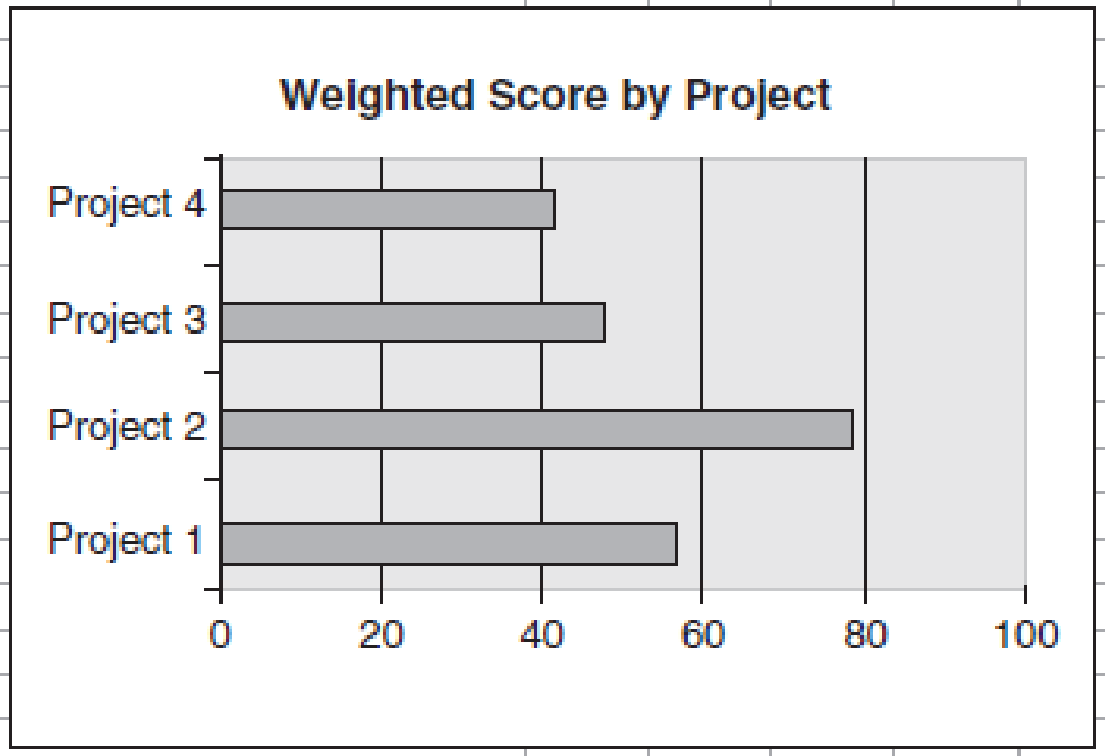
Lifetime ROI = (estimated lifetime benefits – estimated lifetime costs) / estimated lifetime costs

Weighted Score Model – Selecting Projects

- A **weighted scoring model** is a tool that provides a systematic process for selecting projects based on many criteria.
- These criteria can include factors such as meeting broad organizational needs; addressing problems, opportunities, or directives; the amount of time it will take to complete the project; the overall priority of the project; and projected financial performance Of the project.

Example- Weighted Score Model

Criteria	Weight	Project 1	Project 2	Project 3	Project 4
Supports key business objectives	25%	90	90	50	20
Has strong internal sponsor	15%	70	90	50	20
Has strong customer support	15%	50	90	50	20
Uses realistic level of technology	10%	25	90	50	70
Can be implemented in one year or less	5%	20	20	50	90
Provides positive NPV	20%	50	70	50	50
Has low risk in meeting scope, time, and cost goals	10%	20	50	50	90
Weighted Project Scores	**100%**	**56**	**78.5**	**50**	**41.5**



Business Drivers

- Business Driver: The root cause(s) for the project development.
- Business drivers are the crucial forces behind the successful development of a project.
- As the result one can define the business case for sustainable project development.

Critical Success Factors

- Critical success factors (CSFs), also known as Key Results Areas (KRAs), refer to the activities that must be completed to a high standard of quality in order to achieve the goals of your project.
- CSFs are a way to prioritize certain tasks as the project plan is being executed.
- Having clear CSFs helps the project team clarify what needs to be worked on first or needs special attention, allowing people to work together to achieve the project's main objectives.

Feasibility Study or Project Justification – Business Case

- Introduction & Background to the proposal
- The market
- Organization & Operational Infrastructure
- Benefits
- Outline implementation Plan
- Costs
- The financial case
- Risks

END OF TOPIC 3

-COMING UP!!!!!!

-Project Team

-Project Scope Management

-WBS